

The Reaction of Organozinc Reagents with Trifluoroacylated Phosphonates: Synthesis of Trifluoromethylated α,β -Unsaturated Esters with an Active Methylene Moiety

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Received 6 October 2003; revised 4 February 2004; accepted 18 February 2004

ABSTRACT: *The consecutive reaction of phosphonates **1** with *n*-butyllithium, trifluoroacetic anhydride, and organozinc compounds gives trifluoromethylated α,β -unsaturated esters with an active methylene moiety and predominant *Z*-selectivity in 54–78% (3 steps) yields. © 2004 Wiley Periodicals, Inc. Heteroatom Chem 15:289–292, 2004; Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/hc.20018*

INTRODUCTION

Much attention has been devoted to the synthesis of α,β -unsaturated acids and their esters, because such compounds have been noted as important structure feature in a number of naturally occurring compounds which show biological activities [1]. They are useful intermediates in synthesis of some bioactive compounds [2] and are able to undergo many useful synthetic transformations [3]. In the past several decades fluorinated α,β -unsaturated esters have attracted much interest, particularly for

the synthesis of fluorine-containing bioactive compounds [4]. Recently, stereoselective synthesis of trisubstituted and tetrasubstituted α,β -unsaturated esters via copper-catalyzed coupling of enol triflates of β -ketoesters with Grignard-based zinc ate complexes was reported [5]. However, to the best of our knowledge, the synthesis of fluorinated tetrasubstituted α,β -unsaturated esters is still limited [6].

RESULTS AND DISCUSSION

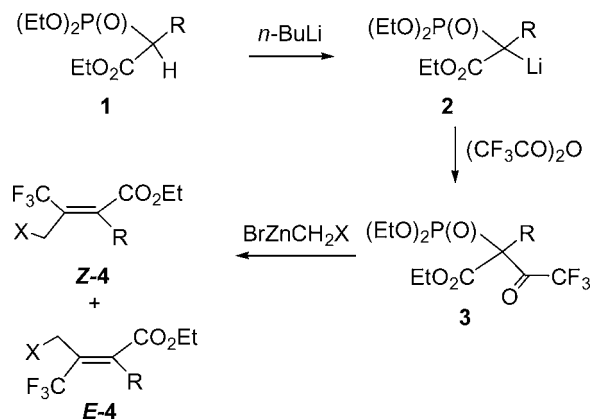
As part of our continuing investigation of synthetic application of consecutive reaction of phosphorus compounds in organic synthesis [6b,7] herein we report the consecutive reaction of phosphonates **1** with *n*-butyllithium, trifluoroacetic anhydride, and organozinc compounds to give trifluoromethylated α,β -unsaturated esters with an active methylene moiety and predominant *Z*-selectivity. The reaction sequence is shown in Scheme 1.

Phosphonate **1** was treated with *n*-butyllithium in tetrahydrofuran (THF) at -78°C and the resulting carbanion **2** reacted with trifluoroacetic anhydride to form the intermediate **3** which was further reacted with organozinc compounds, followed by elimination of phosphate anion, giving substituted trifluoromethylated α,β -unsaturated esters with predominant *Z*-selectivity in 54–78% yields (3 steps). The results are summarized in Table 1.

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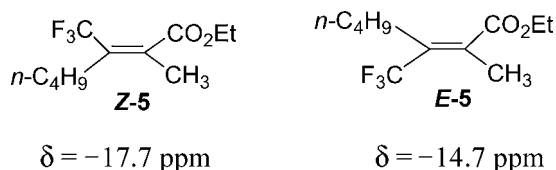
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SCHEME 1

The ^{19}F NMR data of similar compounds **Z-5** and **E-5** have been reported [6b]:



If the trifluoromethyl group is *cis* with respect to the ester group, the chemical shifts of the trifluoromethyl group appear downfield (**Z-5**), while those *trans* with respect to the ester group are shifted upfield (**E-5**). In the cases of **4a–4e**, the chemical shifts of trifluoromethyl group are in the range of -17.0 to -18.1 ppm, hence the configurations of the **4a–4e** could be ascertained as *Z*-isomers. In addition, on the basis of coupling constant across the double bond reported in the literature [8], if the trifluoromethyl group was *trans* with respect to the F group,

TABLE 1 Substituted Trifluoromethylated α,β -Unsaturated Esters Prepared

Compound	R	X	Yield (%) ^a	Ratio (Z:E) ^b
4a	CH ₃	CO ₂ Me	66	100:0
4b	CH ₃	CO ₂ Et	62	100:0
4c	CH ₃	CO ₂ Pr ⁱ	54	100:0
4d	CH ₃	CO ₂ Bu ^t	62	100:0
4e	CH ₃	CH ₂ =CH	57	100:0
4f	F	CO ₂ Et	74	5:95 ^c
4g	F	CO ₂ Bu ^t	78	34:66 ^c
4h	F	CH ₂ =CH	66	14:86 ^c

^aIsolated yields.

^bThe ratios of *E*- and *Z*-isomers are estimated on the basis of NMR data.

^cAccording to the sequence rules, in **4f–4h** (fluorine group instead of methyl group), when the fluorine group is *trans* with respect to the trifluoromethyl group, the stereoisomer is assigned as the *E*-isomer and, conversely in **4a–4e** they are assigned as the *Z*-isomer.

the $^4J_{\text{FFtrans}}$ ranged from 7 to 13 Hz, while for those *cis* with respect to the F group, the $^4J_{\text{FFcis}}$ ranged from 21 to 31 Hz. Thus the configuration of **4f–4g** could be ascertained.

It is noteworthy that the title compounds with an active methylene moiety are able to deprotonate by a base forming a carbanion, which would be a potential useful fluorine-containing building block in organic synthesis.

When the organozinc reagent with double bond moiety ($\text{BrZnCH}_2\text{CH}=\text{CH}_2$) (**4e** and **4h**) is used as starting material, the trifluoromethylated dienyl carboxylates were obtained.

In summary this methodology provides a convenient synthesis of the title compounds, which would be useful intermediates in the synthesis of fluorine-containing biological active compounds.

EXPERIMENTAL

All boiling points are uncorrected. The IR spectra of liquid products were determined as films on a Digilab FTS-20E spectrometer. ^1H NMR spectra were recorded on a Bruker AM-300 (300 MHz) spectrometer (values in ppm from SiMe_4 , in CDCl_3 ; J values are given in Hz). ^{19}F NMR spectra were taken on a Varian EM-360 (60 MHz) spectrometer with $\text{CF}_3\text{CO}_2\text{H}$ as external standard, positive for upfield shifts. Mass spectra were measured on a Finnigan GC-MS-4021 mass spectrometer. HRMS data were obtained on a Finnigan-Mat 8430 high resolution mass spectrometer.

Diethyl (1-ethoxycarbonyl-1-fluoromethyl)-phosphonate

It was prepared according to the known method [9].

General Procedure for the Preparation of Substituted Trifluoromethylated α,β -Unsaturated Esters

Treatment of phosphonate (**1**) (3 mmol) with *n*-butyllithium (3 mmol) in absolute THF (15 ml) at -78°C under nitrogen gave the phosphoryl-stabilized carbanion **2**, which was stirred at -78°C for 0.5 h under nitrogen. Trifluoroacetic anhydride (0.63 g, 3 mmol) was added to it in one portion. After stirring at -78°C for 0.5 h, the reaction mixture was allowed to warm to 20°C and the organozinc reagent [9 mmol, prepared from 9 mmol of bromoacetic ester and 9 mmol (0.59 g) of zinc powder] [10] was added to the mixture, which was stirred for another 4 h. The reaction mixture was poured into 0.5 N HCl solution (15 ml) and the water layer was

extracted with ethyl ether (3 $\times$ 20 ml). The combined organic layer was washed with water (3 $\times$ 15 ml) and dried over Na_2SO_4 . Evaporation of the solvent gave a residue, which was purified by column chromatography eluting with petroleum ether (60–90°C)-ethyl acetate (99:2) to give the product **4**.

Z-Ethyl 4-methoxycarbonyl-3-trifluoromethyl-2-methylbut-2-enoate (**Z-4a**)

Yield: 66%; bp. 78°C/2 mm Hg. *Z:E* = 100:0. IR (neat): ν = 2990, 1750, 1730, 1660, 1440, 1270, 1130 cm^{-1} . ^1H NMR (CDCl_3/TMS): δ = 4.24 (q, J = 7.1 Hz, 2H), 3.70 (s, 3H), 3.48 (s, 2H), 2.14 (q, J = 1.8 Hz, 3H), 1.30 (t, J = 7.1 Hz, 3H). ^{19}F NMR (CDCl_3/TFA): δ = –17.2 (s, 3F). MS: m/z (%) = 255 (M^+ + 1, 18), 223 (87), 209 (100), 208 (50), 195 (30), 181 (86), 180 (66), 167 (38). Anal Calc. for $\text{C}_{10}\text{H}_{13}\text{F}_3\text{O}_4$ (254.20): C, 47.25; H, 5.15. Found: C, 46.85; H, 4.99.

Z-Ethyl 4-ethoxycarbonyl-3-trifluoromethyl-2-methylbut-2-enoate (**Z-4b**)

Yield: 62%; bp. 89°C/2 mm Hg. *Z:E* = 100:0. IR (neat): ν = 2990, 1750, 1730, 1660, 1270, 1200, 1130, 1040 cm^{-1} . ^1H NMR (CDCl_3/TMS): δ = 4.24 (q, J = 7.1 Hz, 2H), 4.16 (q, J = 7.1 Hz, 2H), 3.46 (s, 2H), 2.34 (q, J = 2.1 Hz, 3H), 1.30 (t, J = 7.1 Hz, 3H), 1.25 (t, J = 7.1 Hz, 3H). ^{19}F NMR (CDCl_3/TFA): δ = –17.0 (s, 3F). MS: m/z (%) = 269 (M^+ + 1, 8), 223 (100), 222 (38), 195 (82), 194 (31), 167 (83), 166 (38), 148 (36), 128 (34). Anal Calc. for $\text{C}_{11}\text{H}_{15}\text{F}_3\text{O}_4$ (268.23): C, 49.26; H, 5.64. Found: C, 48.92; H, 5.67.

Z-Ethyl 4-*i*-propoxycarbonyl-3-trifluoromethyl-2-methylbut-2-enoate (**Z-4c**)

Yield: 54%; bp. 90°C/2 mm Hg. *Z:E* = 100:0. IR (neat): ν = 2990, 1740, 1660, 1270, 1200, 1130, 1040 cm^{-1} . ^1H NMR (CDCl_3/TMS): δ = 5.01 (sept, J = 6.3 Hz, 1H), 4.24 (q, J = 7.1 Hz, 2H), 3.43 (s, 2H), 2.13 (q, J = 1.9 Hz, 3H), 1.30 (t, J = 7.1 Hz, 3H), 1.22 (d, J = 6.3 Hz, 6H). ^{19}F NMR (CDCl_3/TFA): δ = –17.3 (s, 3F). MS: m/z (%) = 283 (M^+ + 1, 23), 241 (46), 223 (100), 196 (9), 195 (38), 167 (14), 43 (25). Anal Calc. for $\text{C}_{12}\text{H}_{17}\text{F}_3\text{O}_4$ (282.25): C, 51.06; H, 6.07. Found: C, 50.86; H, 5.91.

Z-Ethyl 4-*t*-butoxycarbonyl-3-trifluoromethyl-2-methylbut-2-enoate (**Z-4d**)

Yield: 62%; bp. 96°C/2 mm Hg. *Z:E* = 100:0. IR (neat): ν = 2990, 1740, 1660, 1370, 1130, 1040 cm^{-1} . ^1H NMR (CDCl_3/TMS): δ = 4.24 (q, J = 7.1 Hz, 2H), 3.36 (s, 2H), 2.12 (q, J = 2.2 Hz, 3H), 1.40 (s, 9H),

1.30 (t, J = 7.1 Hz, 3H). ^{19}F NMR (CDCl_3/TFA): δ = –17.5 (s, 3F). MS: m/z (%) = 297 (M^+ + 1, 12), 242 (13), 241 (100), 224 (5), 223 (45), 195 (7), 175 (5).

HRMS: m/z Calc. for $\text{C}_{13}\text{H}_{19}\text{F}_3\text{O}_4$: 296.1235, Found 296.1237.

Z-Ethyl 3-trifluoromethyl-2-methylhexa-2,5-dienoate (**Z-4e**)

Yield: 57%; bp. 85°C/10 mm Hg. *Z:E* = 100:0. IR (neat): ν = 2990, 1730, 1730, 1670, 1270, 1190, 1130 cm^{-1} . ^1H NMR (CDCl_3/TMS): δ = 5.88–5.72 (m, 1H), 5.16–5.09 (m, 2H), 4.26 (q, J = 7.1 Hz, 2H), 3.06 (d, J = 6.0 Hz, 2H), 2.09 (q, J = 2.4 Hz, 3H), 1.32 (t, J = 7.1 Hz, 3H). ^{19}F NMR (CDCl_3/TFA): δ = –18.1 (s, 3F). MS: m/z (%) = 220 (M^+ – 2, 3), 179 (31), 177 (44), 149 (56), 130 (53), 129 (100), 109 (31), 79 (53). Anal Calc. for $\text{C}_{10}\text{H}_{13}\text{F}_3\text{O}_2$ (222.20): C, 54.05; H, 5.90. Found: C, 53.76; H, 5.89.

Ethyl 4-ethoxycarbonyl-3-trifluoromethyl-2-fluorobut-2-enoate (**4f**)

Yield: 74%; bp. 86°C/2 mm Hg. *Z:E* = 5:95. IR (neat): ν = 2990, 1750, 1660, 1320, 1200, 1160, 1090, 1020 cm^{-1} . ^1H NMR (CDCl_3/TMS): δ = 4.36 (q, J = 7.1 Hz, 2H), 4.19 (q, J = 7.1 Hz, 2H), 3.72 (d, J = 1.6 Hz, 0.05 $\times$ 2H), 3.40 (d, J = 3.8 Hz, 0.95 $\times$ 2H), 1.35 (t, J = 7.1 Hz, 3H), 1.26 (t, J = 7.1 Hz, 3H). ^{19}F NMR (CDCl_3/TFA): δ = –17.5 (d, J = 10 Hz, 0.95 $\times$ 3F), –14.5 (d, J = 24 Hz, 0.05 $\times$ 3F), 28.8 (m, 1F). MS: m/z (%) = 273 (M^+ + 1, 9), 253 (13), 227 (26), 199 (43), 172 (86), 171 (100), 152 (60), 132 (68). Anal Calc. for $\text{C}_{10}\text{H}_{12}\text{F}_4\text{O}_4$ (272.19): C, 44.13; H, 4.44. Found: C, 43.91; H, 4.48.

Ethyl 4-*tert*-butoxycarbonyl-3-trifluoromethyl-2-fluorobut-2-enoate (**4g**)

Yield: 78%; bp. 88°C/2 mm Hg. *Z:E* = 34:66. IR (neat): ν = 2990, 1740, 1660, 1380, 1330, 1280, 1160, 1020 cm^{-1} . ^1H NMR (CDCl_3/TMS): δ = 4.42–4.34 (m, 2H), 3.66 (d, J = 1.7 Hz, 0.34 $\times$ 2H), 3.34 (d, J = 3.8 Hz, 0.66 $\times$ 2H), 1.47 (s, 9H), 1.37 (t, J = 7.2 Hz, 3H). ^{19}F NMR (CDCl_3/TFA): δ = –17.8 (d, J = 10 Hz, 0.66 $\times$ 3F), –14.0 (d, J = 24 Hz, 0.34 $\times$ 3F), 29.8 (m, 0.66 $\times$ 1F), 32.4 (m, 0.34 $\times$ 1F). MS: m/z (%) = 301 (M^+ + 1, 16), 245 (100), 227 (28), 57 (54). HRMS: m/z Calc. for $\text{C}_{12}\text{H}_{16}\text{F}_4\text{O}_4$: 300.0985, Found 300.0953.

Ethyl 3-trifluoromethyl-2-fluorohexa-2,5-dienoate (**4h**)

Yield: 66%; bp. 82°C/10 mm Hg. *Z:E* = 14:86. IR (neat): ν = 2990, 1740, 1670, 1640, 1270, 1190, 1380,

1350, 1280, 1220, 1140 cm^{-1} . ^1H NMR (CDCl_3/TMS): $\delta = 5.90\text{--}5.72$ (m, 1H), $5.25\text{--}5.10$ (m, 2H), 4.35 (q, $J = 7.1$ Hz, 2H), 3.40 (d, $J = 6.1$ Hz, $0.86 \times 2\text{H}$), 3.10 (dd, $J = 6.1, 3.6$ Hz, $0.14 \times 2\text{H}$), 1.37 (t, $J = 7.2$ Hz, 3H). ^{19}F NMR (CDCl_3/TFA): $\delta = -17.6$ (d, $J = 10$ Hz, $0.14 \times 3\text{F}$), -15.6 (d, $J = 24$ Hz, $0.86 \times 3\text{F}$), $32.5\text{--}34.8$ (m, 1F). MS: m/z (%) = 227 ($\text{M}^+ + 1$, 28), 199 (34), 172 (47), 171 (73), 152 (38), 132 (37), 57 (100). HRMS: m/z Calc. for $\text{C}_9\text{H}_{10}\text{F}_4\text{O}_2$: 226.0617, Found 226.0604.

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